Various Types of Charts Transcript

Hi, today we will be discussing about different types of charts. We will be focussing on understanding the different types of charts and what are best suited for developing particular visuals. In fact, we'll be understanding what is the purpose of the visualisation you are trying to develop and which chart is best suited for that particular purpose.

It is important that your visualisation contains all the information that you want to present in a coherent fashion and choosing the best chart for the purpose solves the major portion of this problem. With this understanding in the mind, we'll try to understand the common chart types and the purposes for which they are used in different visualisation exercises. The first type of charts that you encounter are the comparison charts.

This is the basic feature of any visualisation exercise where you want to compare categories and present the ranking of different types of variables within those categories to give a sense of how they compare against each other and to drive some story. In our example, we have seen the water turnout ratio across states. It's inherently a comparison exercise trying to show some states are above the national average in comparison to the states which are below the national average.

Some of them are highest ranking as 1, 2, 3 and some states are like Bihar and Uttar Pradesh are at the bottom with lowest water turnout ratio. Inherently when you are doing comparison, you are ranking them and the charts best suited for this purpose are the bar charts. There are two types of bar charts.

One is the horizontal bar chart, another is the vertical bar chart. These are very intuitive presentations where the x-axis contains the category variable that you want to represent and the y-axis is a quantitative variable which tries to capture the quantity of the variable and the length of the bar represents the quantity and you can compare two categories of variables and make a visual comparison very easily. In this example, we are showing on the left side of the slide the Twitter reactions to the airline complaints or the airline surveys.

Some of them are positive, some of them are split a feedback on airline performance, some are positive, some are most of them are negative and neutral. The very fact you are looking at this picture, you can say there are the negative sentiment towards the airline is higher compared to the positive or the neutral sentiment. Okay, this is a very simple bar chart which enables comparison.

You can also use bar charts for other purposes. For example, within the negative sentiment category, you can try to find the source for negative sentiment on the horizontal bar chart. In the earlier slide, it is a vertical bar chart.

The same thing flipped, you can get a horizontal bar chart where you have a lot of categories to present and the x-axis represents the quantitative variable. Here also the comparisons can be made very easily and it is in the descending order following the principle of continuity of this slide. We can see customer service issue is the major cause for negative comments on the Twitter for the airline.

So this kind of decision insights can be easily brought and the company can focus on the customer issues and bring down its negative response or negative feedback from the customers. So as I was saying, there are three types of bar charts, horizontal bar chart, vertical bar chart, primarily two. There are primarily two types of bar charts, horizontal bar chart and vertical bar chart and there is also one more type of bar chart which is called stacked bar chart.

These are different types of bar charts that are available. As we go through the exercises in the course and other lessons, we will go very deep into the types of bar chart and how they are used. For now, you can take in mind whenever there is a comparison exercise, primarily it is done using bar charts and they are the best and simple visualisations that are out there to portray the data.

Then you have distribution. We have done comparison, now we are coming to distribution. Distribution is how particular variable is distributed across a class of data.

For example, height of males, weight of different types of cars. So you can take the smallest car which weighs about 1000 kilogrammes to very high performing car which is about 5000 kilogrammes to 6000 kilogrammes and the interim range is populated by different set of cars. So you can range all, rank all the cars or distribute all the cars according to their weight.

Similarly, you can create a distribution of all mains depending upon their height. So distribution is also a natural form of presenting data where you are trying to understand the normal tendency of the data and the best graphs that are there to do this purpose is the histogram and the scatter plot. We will be taking the example of histogram and what does it help us in doing.

Once we have a distribution, we can see the clusters that are there in the data or subgroups. For example, you might find cars that are below 2000 kgs to be more in number than the cars which are above 5000 kgs. Though there might be a distribution of cars in between, you know mid-segment cars are most popular with the retail commuters.

So you have more number of models in that range. So this kind of insights can be brought from a distribution graph which is given by a histogram. In this case, we also look at the Twitter complaints of the airline and see the number of words in the text.

So we have the number of word in the text and count of number of words in the text on the y axis. We are seeing the distribution. So what does the distribution say? For example, there are around 800 tweets.

800 tweets between 20 to 25 words. Now what is the purpose of this distribution? Let us say you are developing an AI based chatbot response or to understand the tweets. You need to build the capacity of the system.

You want an average boast of your tweets. So roughly they say there are about 3000 or 5000 tweets that you have analysed. On an average, the tweet is about 20 to 30 words long.

So that is the essence that you are getting from this graph. So the distribution graphs will tell us about the average behaviour along with the distribution so that we can know how to base our decisions. So if you create a text space for 30 words, you should be able to cater to a large number of customer complaints without customers having to write 2 or 3 more messages.

So this is an insight that you can get from this graph that what should be the length of customer texting that you should allow. It is around 20 to 25 words because most average number of customers will fall in that and the ones to the left of that will have less number of words. They can also use this space.

There might be few outliers to whom you can say okay it is fine for them to either call or provide one more message. So it's a business decision being enabled from this visual. So in this way distribution graphs are also very useful to the business insights and they can give us the composition of the data in terms of clusters, where is the average and the statistical properties are.

Then we go to the composition. We have seen comparison, distribution, now composition. When we come to composition we are talking about a particular category.

Let us say the number of total number of marks that a student has gained and the six different subjects that he has studied. So there is always whenever we are talking about composition there is a whole and we are seeing the sum of its parts. So there is one super set or the overarching category under which there are different subcategories and we are seeing how much is the subcategory constituting to the overall category.

Therefore the word composition. For example I got 90% marks do I did I get full marks in one subject or it is equally distributed across all subjects. In some subjects I have failed very badly.

So this kind of questions can be answered by composition. So we can see this composition chart also as a useful tool for driving critical insights. The same Twitter sentiment can be presented in a pie chart which can easily show the negative feedback is very higher and this negative feedback can be further divided into root causes as we have seen in the horizontal bar chart and we can identify the cause of let us say what is the main cause for negative sentiment and what is it shared in the overall sentiments of the company.

So this is something that composition enables us to do. And finally we come to relationship. A relationship is also a very important aspect of visualisation.

We do a lot of exploratory analysis while doing visualisation. If you remember the most important feature the reason for doing visualisation is one it enables decision-making, two it enables uncover explore hidden patterns and trends in the data and third is telling the argumentation in supporting an argumentation of putting forward a direction to our story. A major part of this is the uncovering hidden patterns.

Sometimes when we have a data set a priory we might not know what are the data patterns in it. So we might have to explore. While exploring certain relationships there are particular graphs that help us in exploring such relationships.

Most famous graph is the scatter plot which we will discuss in detail in later classes. There are other types of relationship plots like bubble chart. There are other important charts like the bubble chart.

See this bubble chart tries to show the relationship between variables of interest along all the qualitative variables. For example the bubble chart captures the relationship between two variables. Let us say in this case again the Twitter feedback is taken into account.

You can see the positive feedback is given in blue, neutral feedback in grey and negative feedback is in brown. Very intuitive colouring and then you can also see the size of the bubble reflects the share or the number of complaints in that particular category. The major category immediately you can identify major sentiment is what negative sentiment and what is the source of the negative sentiment is the customer service related issue.

We can see these things for immediately from the relationship chart. It shows this is the sentiment the negative sentiment is very higher. Again there is a just clad principle operating here principle of common region.

If you see the most the immediately striking feature of the visual is the common region is dominated by brown colour which is reflective of the negative sentiment. So immediately you can draw a conclusion that it is negative sentiment in place and then within that negative region you have a larger space allocated to the customer service issue and you can pinpoint it as the major cause and this is how you can use relationship charts to drive your visualisation. There is similar kind of relationship chart called word cloud.

These days it is very famous because it will it is based on text analysis. The text is collected. Just to give more context to build this bubble chart you need to count two things.

One is you have to first see the nature of complaint and categorise that it into customer service related baggage issue or flight delay. If it is positive good flight service like then count the number of tweets. In this case you have to forcefully categorise them into two or three areas or into one of the buckets.

It might not always be feasible because some tweets might be having both positive or negative role. It's a call that you have to take whether you will place them in a positive category neutral category or a negative category. You might have to design certain rules which might not be fully objective whereas in a word cloud you can count the number of times a particular word is repeated and highlight its impostment.

Let us say bad service bad service bad service. You just leave out what is there before and after the very fact the bad service is there can be treated as a negative sentiment and take such phrases and count them. It is more objective and it shows which is the most important word or most frequently repeated word in the entire feed text and what is the tonality of that word so you can get both the positive negative neutral tone and how many times it has been done in this word count and it intuitively drives once we look customer service issue as the major issue that is bringing down the sentiment of the airline.

So this is a relationship chart so we will be seeing them through different examples throughout the course but again driving home the point whenever you want to build an effective visual you have to choose the right chart. Now we will look at what are the common mistakes that one should avoid while preparing charts. First is too much information on a chart will create clutter and that has to be avoided and unnecessarily plotting of points also should be avoided.

You should always provide labels so that the reader is able to understand what does the x-axis and y-axis convey and you should scale the x and y-axis so that the graphic is clearly conveying the underlying trend. If you do not scale the data the patterns can look distorted and wrong conclusions might be drawn and finally select the chart type clearly so that the audience are able to derive the message that you are trying to get across. Just as a summary when you want to compare certain things you can use a grouped bar chart when you want to compare something over a time line chart and show distribution of two variables scatter plot percentage of sales across regions you can choose tagged bar chart then you have geographical data with both positive and negative values you can use a map then explore relationship between three factors you can use mobile chat.

So we have not covered all of these things but going forward in the course we will be definitely looking at some of these examples both in the classes and in the experiential learning sessions. So we will take a minute to do the quiz so that your concepts are crystallised.

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